# A Unified Geometric Framework for Psychophysical Interaction Dynamics: From Discrete Systems to Riemannian Field Equations with Numerical Simulation

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**Abstract**

We present a unified and self-contained mathematical framework that develops the theory of psychophysical interpersonal interaction across four successive levels of abstraction: discrete dynamical systems, continuous flow, second-order emotional dynamics, and Riemannian geometric field theory. Beginning from the Weber–Fechner psychophysical law, we construct a discrete-time recurrence $I_S(t+1) = \gamma I_S(t)$ governed by the interaction resistivity matrix $\gamma$, provide complete spectral stability analysis via the Jury criterion, and identify bifurcation regimes corresponding to convergence, oscillation, and divergence. We then derive the continuous-time generator equation $\dot{I_S} = M(t)I_S$ and prove the equivalence $\gamma = e^{M\Delta t}$. Differentiating twice, we identify the emotion observable $E(t) = \dot{M}(t) + M(t)^2$ as the second-order curvature of interaction flow, prove that $E(t) = 0 \Leftrightarrow M(t) = (t + \tau_0)^{-1}$, and derive the invariant healing time $T_{\text{heal}}(\epsilon) \sim \mathcal{O}(\epsilon^{-1/2})$. Imposing the psychophysical constraint $I_P = \gamma(t, I_S)I_S$ on the ambient Euclidean space induces a Riemannian metric $G(t, I_S) = I + J^\top J$ on the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$. We derive the Levi-Civita connection, compute the full Ricci tensor decomposition, and lift the three-layer hierarchy (discrete $\to$ continuous $\to$ emotional) into coherent geometric objects. Finally, under axioms of locality, diffeomorphism covariance, and second-order dynamics, we prove that the unique admissible action functional is the Einstein–Hilbert form, yielding the interaction field equation $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$, which governs the geometry of interaction space. The theoretical framework is validated by a numerical simulation of multi-agent geodesic flow on the interaction field manifold, implemented using pre-computed symbolic tensors (SymPy) and high-speed numerical evaluation (NumPy).

**Keywords:** Weber–Fechner law, interaction resistivity matrix, Jury stability, continuous flow generator, emotion dynamics, Riemannian geometry, interaction manifold, Ricci curvature, Einstein field equations, multi-agent simulation.

# Contents

# 1 Introduction

## 1.1 Motivation and Context

Interpersonal interaction represents one of the fundamental dynamical systems in psychological and social science. Despite extensive empirical work, a unified mathematical framework connecting psychophysical principles to interaction dynamics—and ultimately to a geometric field theory—has been absent.

This paper synthesises four successive theoretical contributions by the author into a single, self-contained document. The unifying thread is the Weber–Fechner law [3], which posits logarithmic perceptual scaling and serves as the psychophysical foundation from which all subsequent structure is derived.

At the first level (Section 3), we construct a *discrete-time linear recurrence* governed by the interaction resistivity matrix $\gamma$, provide complete eigenvalue analysis, Jury stability conditions, and bifurcation regimes.

At the second level (Section 4), we derive the *continuous-time interaction flow* $\dot{I}_S = M(t)I_S$ as the infinitesimal limit of the discrete system, prove the generator equivalence $\gamma = e^{M\Delta t}$, and characterise stability and oscillation via spectral theory and Hopf-type bifurcations.

At the third level (Section 5), we differentiate the flow equation a second time to identify *emotion* as the operator $E(t) = \dot{M}(t) + M(t)^2$, prove the unique relaxation condition, and derive the invariant healing time.

At the fourth and highest level (Sections 6–8), we impose the psychophysical constraint on an ambient Euclidean space to induce a *Riemannian metric* on the interaction state space, derive the full Levi-Civita connection and Ricci curvature, and prove—under minimal axioms—that the unique governing action yields *Einstein-type interaction field equations*.

Finally, in Section 9, we present a *numerical simulation* of multi-agent dynamics on the interaction field manifold, using symbolic pre-computation of the Christoffel symbols and Einstein tensor followed by high-speed numerical evaluation of geodesic forces.

## 1.2 Contributions

The principal contributions of this unified paper are:

(i) Complete discrete-time analysis: eigenvalue spectra, Jury criterion, and spectral bifurcation theory for the interaction resistivity system.

(ii) Derivation of the continuous flow generator and proof of $\gamma = e^{M\Delta t}$.

(iii) Definition and characterisation of the emotion observable $E(t) = \dot{M} + M^2$, including the unique relaxation solution and healing time.

(iv) Construction of the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ with time-dependent and state-dependent Riemannian metrics.

(v) Complete Levi-Civita connection and Ricci tensor decomposition into temporal and spatial contributions.

(vi) A three-layer geometric lifting theorem unifying discrete, continuous, and emotional dynamics within the manifold.

(vii) Uniqueness proof of the Einstein–Hilbert interaction action and derivation of the interaction field equation.

(viii) Numerical simulation validating the geometric field theory via pre-computed symbolic tensors and multi-agent geodesic flow.

# 2 Psychophysical Foundations

## 2.1 The Weber–Fechner Law

**Definition 2.1** (Weber–Fechner Encoding)**.** The Weber–Fechner law [3] states that the perceived intensity $P$ associated with a physical stimulus of magnitude $S > 0$ is

$$P = \alpha \ln\left(\frac{S}{S_0}\right), \tag{1}$$

where $\alpha > 0$ is a perceptual sensitivity constant and $S_0 > 0$ is the baseline detection threshold.

Equation (1) follows from the Weber differential law $dP = \alpha\, dS/S$, which encodes diminishing marginal returns: a doubling of stimulus yields only an additive increment in subjective perception.

**Assumption 2.2** (Restricted Psychophysical Generalisation)**.** Within moderate-intensity regimes, social-emotional stimuli obey logarithmic perceptual scaling:

$$P_{\text{social}} \approx \alpha \ln\left(\frac{S_{\text{social}}}{S_0}\right).$$

This is adopted as a local approximation, not a universal psychological law. Extreme affective states may require Stevens' power law or other models.

Three independent lines of evidence support Assumption 2.2: *(i)* Neurobiological: Berkovich et al. [2] demonstrated logarithmic scaling of pleasant emotional responses with stimulus intensity. *(ii)* Decision-theoretic: Prospect Theory [9] and Namboodiri et al. [12] confirm Weber scaling of subjective value. *(iii)* Theoretical parsimony: social interaction fundamentally involves perception–response coupling, suggesting the same psychophysical mechanisms apply.

## 2.2 Interaction Variables and the Resistivity Matrix

For a system of $n$ interacting agents, we distinguish two classes of interaction variables.

**Definition 2.3** (Interaction Variables)**.** The *active interaction vector* $I_S(t) \in \mathbb{R}^n$ represents the expressive signal generated by the agents, with $[I_S(t)]_i$ quantifying agent $i$'s expressive output at time $t$. The *passive interaction vector* $I_P(t) \in \mathbb{R}^n$ represents the perceptual states, with $[I_P(t)]_i$ quantifying agent $i$'s received perception.

Using Weber–Fechner encoding for both reception and expression:

$$[I_P(t)]_i = k_{P,i}\, \alpha_i \ln\left(\frac{S_i(t)}{S_0}\right), \qquad [I_S(t)]_i = k_{S,i} \ln\left(\frac{S_i(t)}{S_0}\right). \tag{2}$$

**Theorem 2.4** (Interaction Resistivity Matrix)**.** *There exists a constant matrix* $\gamma \in \mathbb{R}^{n\times n}$*, the* interaction resistivity matrix*, such that*

$$I_P(t) = \gamma I_S(t), \qquad \gamma_{ij} = \frac{k_{P,i}\, \alpha_i}{k_{S,j}}. \tag{3}$$

*Proof.* From the encoding maps, $[I_P(t)]_i/[I_S(t)]_j = k_{P,i}\alpha_i/k_{S,j}$. Setting $\gamma_{ij} := k_{P,i}\alpha_i/k_{S,j}$ and summing over agents yields (3). □

The diagonal elements $\gamma_{ii}$ represent self-perception feedback loops; off-diagonal elements $\gamma_{ij}$ encode cross-agent perceptual influence.

# 3 Discrete-Time Interaction Dynamics

## 3.1 The Discrete Interaction System

Interpersonal interaction unfolds through repeated perception–action cycles. We represent this as a discrete-time recurrence.

**Definition 3.1** (Discrete Interaction Dynamics)**.** The active interaction state evolves according to

$$I_S(t+1) = \gamma I_S(t), \qquad t \in \mathbb{N}. \tag{4}$$

**Theorem 3.2** (Existence and Uniqueness)**.** *For any* $I_S(0) \in \mathbb{R}^n$*, the recurrence* (4) *has the unique solution*

$$I_S(t) = \gamma^t I_S(0).$$

## 3.2 Stability Analysis: Spectral Radius Criterion

**Proposition 3.3** (Asymptotic Stability)**.** *The equilibrium* $I_S = 0$ *of* (4) *is asymptotically stable if and only if* $\rho(\gamma) < 1$*, where* $\rho(\gamma) = \max_i |\lambda_i(\gamma)|$*.*

## 3.3 Dyadic-Agent Specialisation and Jury Criterion

For $n = 2$, the characteristic polynomial is

$$p(\lambda) = \lambda^2 - \operatorname{tr}(\gamma)\lambda + \det(\gamma),$$

with eigenvalues

$$\lambda_\pm = \frac{\operatorname{tr}(\gamma) \pm \sqrt{\operatorname{tr}(\gamma)^2 - 4\det(\gamma)}}{2}. \tag{5}$$

**Theorem 3.4** (Jury Stability Criterion). *The equilibrium $I_S = 0$ is asymptotically stable if and only if:*

*(i)* $|\det(\gamma)| < 1$,

*(ii)* $1 + \mathrm{tr}(\gamma) + \det(\gamma) > 0$,

*(iii)* $1 - \mathrm{tr}(\gamma) + \det(\gamma) > 0$.

### 3.4 Oscillatory Regimes and Bifurcation Theory

Define the discriminant $\Delta = \mathrm{tr}(\gamma)^2 - 4\det(\gamma)$ and conflict parameter $C = -\gamma_{AB}\gamma_{BA}$.
**Theorem 3.5** (Oscillation Criterion). *The discrete system exhibits oscillatory dynamics if and only if*

$$C > \frac{(\gamma_{AA} - \gamma_{BB})^2}{4}.$$

**Proposition 3.6** (Eigenvalue Phase Transition). *Let* $\gamma(\mu) = \begin{pmatrix} \gamma_{AA} & \mu \\ -\mu & \gamma_{BB} \end{pmatrix}$. *Eigenvalues are real for $|\mu| < \mu^* := |\gamma_{AA} - \gamma_{BB}|/2$ and form a complex conjugate pair for $|\mu| > \mu^*$. A Neimark–Sacker-type transition at $\mu_c$ produces cyclical emotional patterns as a structural consequence of perception–action coupling.*

Table 1 summarises the qualitative phase regimes.

Table 1: Qualitative phase regimes for the discrete interaction system.

| $\rho(\gamma)$ | $\Delta$ | Behavioural Regime |
|---|---|---|
| $< 1$ | $> 0$ | Monotonic convergence (de-escalation) |
| $< 1$ | $< 0$ | Oscillatory convergence (dampened cycles) |
| $= 1$ | any | Neutral equilibrium / sustained oscillation |
| $> 1$ | $> 0$ | Monotonic divergence (escalation) |
| $> 1$ | $< 0$ | Oscillatory divergence (escalating cycles) |

# 4 Continuous-Time Extension and the Generator

## 4.1 The Interaction Flow Matrix

Letting the time step $\Delta t \to 0$, we obtain the continuous flow.
**Definition 4.1** (Interaction Flow Matrix). The *interaction flow matrix* (generator) is

$$M := \lim_{\Delta t \to 0} \frac{\gamma(\Delta t) - I}{\Delta t}, \tag{6}$$

provided the limit exists.
**Theorem 4.2** (Existence and Uniqueness of Continuous Dynamics). *If $M$ is bounded, then*

$$\frac{dI_S}{dt} = M(t) I_S(t) \tag{7}$$

*has the unique solution* $I_S(t) = e^{Mt} I_S(0)$.

## 4.2 Generator–Matrix Equivalence

**Theorem 4.3** (Generator Equivalence)**.** *For any step size* $\Delta t > 0$*:*

$$\gamma(\Delta t) = e^{M\Delta t}, \qquad M = \frac{1}{\Delta t}\log(\gamma). \tag{8}$$

*Proof.* Iterating the continuous solution over $t = k\Delta t$ gives $I_S(k\Delta t) = e^{Mk\Delta t} I_S(0) = (e^{M\Delta t})^k I_S(0)$. Equating with $I_S(k\Delta t) = \gamma^k I_S(0)$ yields $\gamma = e^{M\Delta t}$. □

## 4.3 Continuous-Time Stability and Bifurcation

**Theorem 4.4** (Continuous Stability Criterion)**.** *The equilibrium* $I_S = 0$ *of* (7) *is asymptotically stable if and only if* $\mathrm{Re}(\lambda_i(M)) < 0$ *for all* $i$.

**Theorem 4.5** (Oscillatory Criterion)**.** *For* $n = 2$*, the continuous system exhibits oscillatory trajectories if and only if*

$$C_M := -m_{AB}m_{BA} > \frac{(m_{AA} - m_{BB})^2}{4}.$$

Define qualitative regimes by the spectral properties of $M$:

Table 2: Continuous-time interaction phase regimes.

| $\mathrm{Re}(\lambda)$ | $\mathrm{Im}(\lambda)$ | Interpretation |
|---|---|---|
| $< 0$ | $= 0$ | Monotonic de-escalation |
| $> 0$ | $= 0$ | Monotonic escalation |
| $= 0$ | $= 0$ | Emotional inertia |
| any | $\neq 0$ | Oscillatory cycling |
| $> 0$ | $\neq 0$ | Escalating fight–recovery loops |

## 4.4 Unification with Classical Behavioural Models

**Gottman Marital Interaction Model** [5]**.** The linearised Gottman model $\begin{pmatrix} W(t+1) \\ H(t+1) \end{pmatrix} = \begin{pmatrix} r_1 & c_{HW} \\ c_{WH} & r_2 \end{pmatrix} \begin{pmatrix} W(t) \\ H(t) \end{pmatrix}$ is a special case of (4) with $\gamma_{AA} = r_1$, $\gamma_{BB} = r_2$, $\gamma_{AB} = c_{HW}$, $\gamma_{BA} = c_{WH}$. The continuous extension yields $M = \frac{1}{\Delta t}\log(\gamma)$, establishing the Gottman model as a discrete projection of the psychophysical flow.

**Thorndike Learning Curve** [19]**.** The escape time $t_S(n)$ satisfies $dt_S/dn = -AM^{-1}(t_S)$. Setting $\|M(t_S)\| = c/t_S$ yields

$$t_S(n) = t_S(0)\exp\left(-\frac{A}{c}n\right), \tag{9}$$

recovering the exponential learning decay as a mathematical consequence of the inverse flow matrix.

# 5 Emotion as Second-Order Interaction Dynamics

## 5.1 The Equation of Emotion

When $M(t)$ depends on time, differentiating $\dot{I}_S = M(t)I_S$ gives:

$$\ddot{I}_S = \dot{M}(t)I_S + M(t)\dot{I}_S = \dot{M}(t)I_S + M(t)^2 I_S = \big(\dot{M}(t) + M(t)^2\big)I_S. \tag{10}$$

**Definition 5.1** (Emotion Observable)**.** The *emotion observable matrix* is

$$E(t) := \dot{M}(t) + M(t)^2. \tag{11}$$

**Theorem 5.2** (Equation of Emotion)**.**

$$\ddot{I}_S(t) = E(t)I_S(t). \tag{12}$$

*Emotion is the second-order curvature of interaction flow.*

## 5.2 Emotional Relaxation: Unique Solution

**Definition 5.3** (Emotional Relaxation)**.** The system is in emotional relaxation when $E(t) = 0$, i.e., when $\dot{M}(t) + M(t)^2 = 0$.

For diagonal $M(t) = \mathrm{diag}(m_{11}(t), \ldots, m_{nn}(t))$, the relaxation condition decouples into scalar Riccati equations $\dot{m}_{ii} + m_{ii}^2 = 0$.

**Theorem 5.4** (Unique Relaxation Solution)**.** *The unique maximal solution of $\dot{m}_{ii} = -m_{ii}^2$ with $m_{ii}(0) = m_{i,0} \neq 0$ is*

$$m_{ii}(t) = \frac{1}{t + \tau_i}, \qquad \tau_i := \frac{1}{m_{i,0}}. \tag{13}$$

*Proof.* Separating variables: $dm_{ii}/m_{ii}^2 = -dt$, integrating gives $-1/m_{ii}(t) = -t + C_i$, so $m_{ii}(t) = (t + \tau_i)^{-1}$. Applying $m_{ii}(0) = m_{i,0}$ yields $\tau_i = 1/m_{i,0}$. □

Under relaxation, $M(t) \to 0$ monotonically, and each component evolves as $I_{S,i}(t) = C_i(t + \tau_i)$, representing a gradual unbinding of interaction intensity—mathematically analogous to meditative emotional detachment.

## 5.3 Invariant Healing Time

**Definition 5.5.** The *invariant healing time* is $T_{\mathrm{heal}}(\epsilon) := \inf\{T : \|E(t)\| < \epsilon,\ \forall t \geq T\}$.

**Lemma 5.6.** *If $M(t) = (t + \tau)^{-1}I + \delta M(t)$ with $\|\delta M(t)\| \leq C(t + \tau)^{-2}$, then $\|E(t)\| \leq K(t + \tau)^{-2}$ for some $K > 0$, and*

$$T_{\mathrm{heal}}(\epsilon) \sim \mathcal{O}(\epsilon^{-1/2}) - \tau, \qquad \epsilon \to 0. \tag{14}$$

### 5.4 Structure Across Layers

Table 3: Three-layer structure of the interaction equation.

| Layer | Equation | Phenomenon |
|---|---|---|
| Discrete | $I_S(t+1) = \gamma I_S(t)$ | Turn-based social interaction |
| Continuous | $\dot{I}_S = M(t) I_S$ | Real-time affective flow |
| Emotional | $\ddot{I}_S = E(t) I_S$ | Emotion as acceleration |

## 6 The Interaction Manifold

### 6.1 Constructing the Riemannian Metric

The dynamical systems framework treats the interaction state space as flat Euclidean. To capture the geometric structure imposed by the coupling $I_P = \gamma I_S$, we pass to a Riemannian formulation.

Consider the ambient Euclidean space $(t, I_P^1, I_P^2, I_S^1, I_S^2) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2$ with line element

$$ds^2_{\text{amb}} = dt^2 + dI_P^\top dI_P + dI_S^\top dI_S. \tag{15}$$

Imposing $I_P = \gamma(t) I_S$ at fixed $t$ gives $dI_P|_{dt=0} = \gamma(t)\, dI_S$, so
**Definition 6.1** (Time-Dependent Interaction Metric)**.**

$$G(t) = I + \gamma(t)^\top \gamma(t). \tag{16}$$

$G(t)$ is symmetric, positive definite, and smooth in $t$.

### 6.2 The Three-Dimensional Interaction Manifold

**Definition 6.2** (Interaction Manifold)**.**

$$\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}, \tag{17}$$

equipped with block-diagonal metric

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{pmatrix}, \tag{18}$$

where $a > 0$ is a constant temporal scale factor, $\mu, \nu \in \{0, 1, 2\}$, and $x^0 = t$, $(x^1, x^2) = I_S$.

### 6.3 State-Dependent Extension

Many realistic systems require $\gamma = \gamma(t, I_S)$ to capture emotional escalation, biological saturation, and adaptive feedback. The total differential of $I_P = \gamma(t, I_S) I_S$ at fixed $t$ yields the effective Jacobian

**Definition 6.3** (Effective Jacobian)**.**

$$J(t, I_S) := \gamma(t, I_S) + (\nabla_{I_S}\gamma)I_S, \qquad [(\nabla_{I_S}\gamma)I_S]_{ij} = \frac{\partial \gamma_{ij}}{\partial I_S^k} I_S^k. \tag{19}$$

**Definition 6.4** (State-Dependent Interaction Metric)**.**

$$G(t, I_S) = I + J(t, I_S)^\top J(t, I_S). \tag{20}$$

This metric is symmetric, positive definite, and varies across $\mathbb{R}^2_{I_S}$, generating spatial curvature absent in the time-only case.

# 7 Levi-Civita Connection and Ricci Curvature

## 7.1 The Interaction Flow Tensor

**Definition 7.1** (Interaction Flow Tensor)**.**

$$K^i{}_j := \frac{1}{2} G^{ik}\, \partial_t G_{kj}. \tag{21}$$

## 7.2 Levi-Civita Connection

**Theorem 7.2** (Christoffel Symbols)**.** *For the metric* (18)–(20)*, the non-vanishing Christoffel symbols are*

$$\Gamma^0_{ij} = -\frac{1}{2a^2}\, \partial_t G_{ij}, \tag{22}$$

$$\Gamma^i_{0j} = K^i{}_j, \tag{23}$$

$$\Gamma^i_{jk} = \frac{1}{2} G^{il}(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}). \tag{24}$$

*All other components vanish. The spatial symbols* (24) *are non-zero whenever* $\nabla_{I_S}\gamma \neq 0$*.*

## 7.3 Ricci Tensor Decomposition

**Definition 7.3** (Interaction Acceleration Tensor)**.**

$$\mathcal{E}_{ij} := \dot{K}_{ij} + K_{ik}K^k{}_j. \tag{25}$$

**Theorem 7.4** (Temporal Component)**.**

$$R_{00} = -\mathrm{Tr}(\dot{K}) - \mathrm{Tr}(K^2). \tag{26}$$

**Theorem 7.5** (Spatial Components — General Case)**.**

$$R_{ij} = \mathcal{E}_{ij} + K_{ij}\,\mathrm{Tr}(K) + R^{\text{space}}_{ij}, \tag{27}$$

*where* $R^{\text{space}}_{ij}$ *is the Ricci tensor of* $(G_{ij}, \partial_k G_{ij})$ *arising from state-dependent spatial variation. In the time-only case,* $R^{\text{space}}_{ij} = 0$*.*

### 7.4 Geometric Lifting of the Three-Layer Hierarchy

**Theorem 7.6** (Geometric Three-Layer Hierarchy). *On the interaction manifold* $(\mathcal{M}, g)$*:*

*(i)* ***Discrete layer****:* $\gamma$ *defines the spatial metric* $G = I + \gamma^\top\gamma$*; the map* $I_S \mapsto \gamma I_S$ *is a metric-encoded tangent space transformation.*

*(ii)* ***Continuous layer****: The generator* $M(t)$ *governs temporal metric deformation via* $\dot{G} = \gamma^\top(M^\top + M)\gamma$*, and determines the Christoffel symbol* $\Gamma^i_{0j} = K^i{}_j$*.*

*(iii)* ***Emotional layer****:* $E(t) = \dot{M} + M^2$ *is algebraically mirrored by* $\mathcal{E}_{ij} = \dot{K}_{ij} + K_{ik}K^k{}_j$*, which appears as the dominant curvature term in* $R_{ij}$*.*

*The generator equivalence* $\gamma = e^{M\Delta t}$ *ensures algebraic consistency across all three layers.*

Table 4: The three-layer interaction hierarchy in dynamical and geometric form.

| Layer | Dynamical Form | Geometric Realisation |
|---|---|---|
| Discrete | $I_S(t+1) = \gamma I_S(t)$ | $G = I + \gamma^\top\gamma$; metric encoding |
| Continuous | $\dot{I}_S = M(t) I_S$ | $K^i{}_j = \frac{1}{2}G^{ik}\dot{G}_{kj}$; Christoffel |
| Emotional | $\ddot{I}_S = E I_S$, $E = \dot{M} + M^2$ | $\mathcal{E}_{ij} = \dot{K}_{ij} + K_{ik}K^k{}_j \subset R_{ij}$ |

### 7.5 Non-Triviality Conditions

**Theorem 7.7.** *If* $\nabla_{I_S}\gamma \neq 0$ *in some open set* $U \subseteq \mathbb{R}^2_{I_S}$*, then* $R^{\text{space}}_{ij}$ *is generically non-zero in* $U$*. The manifold has vanishing Ricci curvature only in the degenerate case* $\gamma = \text{const}$*.*

## 8 Interaction Field Equations

### 8.1 Axioms of the Interaction Action

We seek a governing equation for the interaction geometry via a variational principle.
**Axiom 8.1** (Locality). $S[g] = \int_{\mathcal{M}} \mathcal{L}(g, \partial g, \partial^2 g, \ldots)\, d\mu_g$.
**Axiom 8.2** (Diffeomorphism Covariance). $S[g] = S[\phi^* g]$ for all $\phi \in \mathrm{Diff}(\mathcal{M})$.
**Axiom 8.3** (Metric Dynamics). $g_{\mu\nu}$ is the sole dynamical geometric field.
**Axiom 8.4** (Second-Order Field Equations). The Euler–Lagrange equations contain at most second derivatives of the metric.

### 8.2 Uniqueness of the Interaction Action

**Theorem 8.5** (Uniqueness of the Interaction Action Functional). *Under Axioms 1–4, the unique admissible action on the three-dimensional interaction manifold is, up to boundary terms,*

$$S[g] = \int_{\mathcal{M}} \left( \frac{1}{2\kappa} R + \Lambda \right) \sqrt{|g|}\, d^3x, \tag{28}$$

*where* $R$ *is the Ricci scalar,* $\kappa$ *is the interaction coupling constant, and* $\Lambda$ *is a cosmological-type constant.*

*Proof sketch. (1) Locality and covariance* force the Lagrangian to be built from the Riemann tensor. *(2) In 3D*, the Weyl tensor vanishes; all scalar invariants reduce to functions of $R$ and $R_{\mu\nu}R^{\mu\nu}$. *(3) Second-order constraint* excludes $\nabla R$ and quadratic curvature terms (which generate 4th-order equations). *(4) Admissible scalars*: only 1 (volume) and $R$ (linear in curvature) survive. Setting $\beta = 1/(2\kappa)$ and $\alpha = \Lambda$ gives (28). □

### 8.3 The Interaction Field Equation

Varying (28) with respect to $g^{\mu\nu}$ yields:

$$\boxed{R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu}} \tag{29}$$

This is structurally identical to the Einstein field equations of general relativity, but governs the *geometry of interaction space* rather than spacetime. The geometric dictionary is given in Table 5.

Table 5: Geometric dictionary: interaction space versus spacetime.

| Geometric Object | Interaction Interpretation |
|---|---|
| Metric $G_{ij}(t, I_S)$ | Structure of the interaction space |
| Flow tensor $K^i{}_j$ | Rate of change of interaction coupling |
| Accel. tensor $\mathcal{E}_{ij}$ | Intrinsic interaction acceleration |
| $R_{00}$ | Negative trace of total interaction acceleration |
| $R_{ij}$ | Spatial interaction curvature |
| $T_{\mu\nu}$ | Reinforcement and perceptual sources |
| $\Lambda$ | Baseline interaction intensity |
| $\kappa$ | Interaction coupling constant |

## 9 Numerical Simulation of Interaction Field Dynamics

### 9.1 Overview and Design

To validate the geometric framework, we implement a multi-agent simulation in which agents move on the interaction field manifold under geodesic forces derived from the Einstein tensor. The simulation proceeds in two stages:

1. **Symbolic pre-computation** (Stage 1): Using SymPy, we compute the full Riemannian structure—$\gamma(t, I_S)$, $G(t, I_S)$, Christoffel symbols, Riemann tensor, Ricci tensor, scalar curvature $R$, and Einstein tensor $G_{\mu\nu}$— symbolically, exactly once.

2. **High-speed numerical evaluation** (Stage 2): The symbolic expressions are compiled into NumPy-vectorised functions via `lambdify`, enabling real-time multi-agent dynamics with geodesic forces evaluated on a spatial grid.

## 9.2 Symbolic Tensor Pre-Computation

The interaction matrix $\gamma(t, I_S)$ is defined with Gaussian state-dependence:

$$\gamma_{\text{core}}(x, y) = e^{-(x^2+y^2)/4}, \qquad \gamma(t, I_S) = \begin{pmatrix} \gamma_{\text{core}} & 0 \\ 0 & \gamma_{\text{core}} \end{pmatrix}. \tag{30}$$

The effective Jacobian (19) and state-dependent metric (20) are computed symbolically. The full Christoffel symbol computation:

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} \left( \partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu} \right) \tag{31}$$

is evaluated symbolically over the $3 \times 3 \times 3$ index space. The Riemann tensor, Ricci tensor, and scalar curvature follow in turn, yielding the Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu}. \tag{32}$$

The geodesic force field on the spatial slice is extracted as:

$$F_x = -\partial_x (G_{11} + G_{22}), \qquad F_y = -\partial_y (G_{11} + G_{22}). \tag{33}$$

## 9.3 Multi-Agent Geodesic Dynamics

The interaction source tensor $T_{\mu\nu}$ is modelled as a superposition of agent-centred Gaussian potentials:

$$T_{\text{source}}(x, y) = \sum_{i=1}^{N_{\text{agents}}} 0.7 \exp\left( -\frac{(x - x_i)^2 + (y - y_i)^2}{1.5} \right). \tag{34}$$

The effective field landscape is

$$Z(x, y) = \Lambda - \kappa \cdot T_{\text{source}}(x, y), \tag{35}$$

where the deepest point of $Z$ locates the maximum curvature ("interaction field hole").

Each agent's position is updated via:

$$\begin{pmatrix} x_i(t+1) \\ y_i(t+1) \end{pmatrix} = \begin{pmatrix} x_i(t) \\ y_i(t) \end{pmatrix} + \alpha_f \begin{pmatrix} F_x(x_i - x^*, y_i - y^*) \\ F_y(x_i - x^*, y_i - y^*) \end{pmatrix} + \boldsymbol{\eta}_i, \tag{36}$$

where $(x^*, y^*)$ is the deepest point of the curvature landscape, $\alpha_f = 1.2$ is the force scaling factor, and $\boldsymbol{\eta}_i \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 0.12$ is behavioural noise.

## 9.4 Simulation Parameters

Table 6: Simulation parameters.

| Parameter | Symbol | Value |
|---|---|---|
| Number of agents | $N$ | 6 |
| Spatial domain | – | $[-5, 5]^2$ |
| Grid resolution | – | $25 \times 25$ |
| Geodesic force scale | $\alpha_f$ | 1.2 |
| Noise strength | $\sigma$ | 0.12 |
| Coupling constant | $\kappa$ | 0.6 |
| Cosmological constant | $\Lambda$ | 0.0 |
| Gaussian bandwidth | – | 1.5 |
| Temporal scale factor | $a$ | 1.0 |
| Trajectory memory | – | 25 steps |

## 9.5 Simulation Code

The complete Python implementation is reproduced below, structured to match the two-stage design.

Listing 1: Stage 1: Symbolic tensor pre-computation using SymPy.

```
import numpy as np
import sympy as sp
from sympy import symbols, Matrix, diff, lambdify

t, x, y = symbols('t x y')
coords = [t, x, y]
dim = len(coords)
a_val = 1.0

# Define gamma(t, I_S) with Gaussian state-dependence
gamma_core = sp.exp(-(x**2 + y**2)/4)
gamma_mat = Matrix([[gamma_core, 0], [0, gamma_core]])
I_S = Matrix([x, y])

# Effective Jacobian: J = gamma + (nabla_{I_S} gamma) I_S
dgamma_dx = diff(gamma_mat, x)
dgamma_dy = diff(gamma_mat, y)
grad_gamma_times_IS = Matrix([
    [(dgamma_dx * I_S)[0], (dgamma_dy * I_S)[0]],
    [(dgamma_dx * I_S)[1], (dgamma_dy * I_S)[1]]
])
J = gamma_mat + grad_gamma_times_IS

# State-dependent spatial metric: G = I + J^T J
```

```
G_spatial = sp.eye(2) + J.T * J

# Full 3x3 spacetime metric (block-diagonal)
g_cov = Matrix([
    [a_val**2, 0, 0],
    [0, G_spatial[0,0], G_spatial[0,1]],
    [0, G_spatial[1,0], G_spatial[1,1]]
])
g_inv = g_cov.inv()

# Christoffel symbols: Gamma^lambda_{mu nu}
Christoffel = {}
for lam in range(dim):
    for mu in range(dim):
        for nu in range(dim):
            tmp = 0
            for sigma in range(dim):
                term1 = diff(g_cov[nu, sigma], coords[mu])
                term2 = diff(g_cov[mu, sigma], coords[nu])
                term3 = diff(g_cov[mu, nu], coords[sigma])
                tmp += g_inv[lam, sigma] * (term1 + term2 - term3)
            Christoffel[(lam, mu, nu)] = sp.Rational(1,2) * tmp

# Riemann tensor and Ricci tensor
Riemann = {}
for rho in range(dim):
    for sigma in range(dim):
        for mu in range(dim):
            for nu in range(dim):
                t1 = diff(Christoffel[(rho,sigma,nu)], coords[mu])
                t2 = diff(Christoffel[(rho,sigma,mu)], coords[nu])
                t3 = sum(Christoffel[(rho,mu,l)]*Christoffel[(l,sigma,nu)]
                         for l in range(dim))
                t4 = sum(Christoffel[(rho,nu,l)]*Christoffel[(l,sigma,mu)]
                         for l in range(dim))
                Riemann[(rho,sigma,mu,nu)] = t1 - t2 + t3 - t4

R_ricci = Matrix.zeros(dim, dim)
for mu in range(dim):
    for nu in range(dim):
        R_ricci[mu,nu] = sum(Riemann[(l,mu,l,nu)] for l in range(dim))
```

```
R_scalar = sum(g_inv[mu,nu]*R_ricci[mu,nu]
               for mu in range(dim) for nu in range(dim))

# Einstein tensor G_{mu nu} = R_{mu nu} - (1/2) R g_{mu nu}
Einstein_Tensor = R_ricci - sp.Rational(1,2) * R_scalar * g_cov
R_space_expr = Einstein_Tensor[1,1] + Einstein_Tensor[2,2]

# Geodesic force (negative gradient of curvature landscape)
force_x_expr = -diff(R_space_expr, x)
force_y_expr = -diff(R_space_expr, y)

# Compile to fast NumPy functions
fast_landscape = lambdify((x, y), R_space_expr, 'numpy')
fast_force_x   = lambdify((x, y), force_x_expr, 'numpy')
fast_force_y   = lambdify((x, y), force_y_expr, 'numpy')
```

Listing 2: Stage 2: High-speed numerical multi-agent dynamics.

```
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation

fig = plt.figure(figsize=(14, 6), dpi=75)
ax3d = fig.add_subplot(121, projection='3d')
ax2d = fig.add_subplot(122)

X, Y = np.meshgrid(np.linspace(-5,5,25), np.linspace(-5,5,25))
num_agents = 6;  noise_strength = 0.12;  kappa = 0.6;  Lambda =
    0.0

trajectories_x = [[np.random.uniform(-3,3)] for _ in range(
    num_agents)]
trajectories_y = [[np.random.uniform(-3,3)] for _ in range(
    num_agents)]

def update(frame):
    ax3d.clear();  ax2d.clear()
    current_x = np.array([tx[-1] for tx in trajectories_x])
    current_y = np.array([ty[-1] for ty in trajectories_y])

    # Compute interaction source tensor T_{mu nu}
    T_source = sum(0.7*np.exp(-((X-cx)**2+(Y-cy)**2)/1.5)
                   for cx,cy in zip(current_x, current_y))
    Z = Lambda - kappa * T_source
    min_idx = np.unravel_index(np.argmin(Z), Z.shape)
    deepest_x, deepest_y = X[min_idx], Y[min_idx]

    for i in range(num_agents):
        px, py = trajectories_x[i][-1], trajectories_y[i][-1]
        # Evaluate geodesic force at relative coordinates
```

```
        fx = fast_force_x(px - deepest_x, py - deepest_y)
        fy = fast_force_y(px - deepest_x, py - deepest_y)
        # Update rule (eq. 15): geodesic force + behavioural noise
        next_x = np.clip(px + 1.2*fx + np.random.normal(0, noise_strength),
                         -4.9, 4.9)
        next_y = np.clip(py + 1.2*fy + np.random.normal(0, noise_strength),
                         -4.9, 4.9)
        trajectories_x[i].append(next_x)
        trajectories_y[i].append(next_y)
        if len(trajectories_x[i]) > 25:
            trajectories_x[i].pop(0);  trajectories_y[i].pop(0)

    ax3d.plot_surface(X, Y, Z, cmap='viridis', alpha=0.45)
    ax2d.contourf(X, Y, Z, cmap='viridis', alpha=0.35)

ani = FuncAnimation(fig, update, frames=200, interval=40, repeat=False)
ani.save("interaction_field_simulation.gif", writer='pillow', fps=20)
```

## 9.6 Expected Results and Interpretation

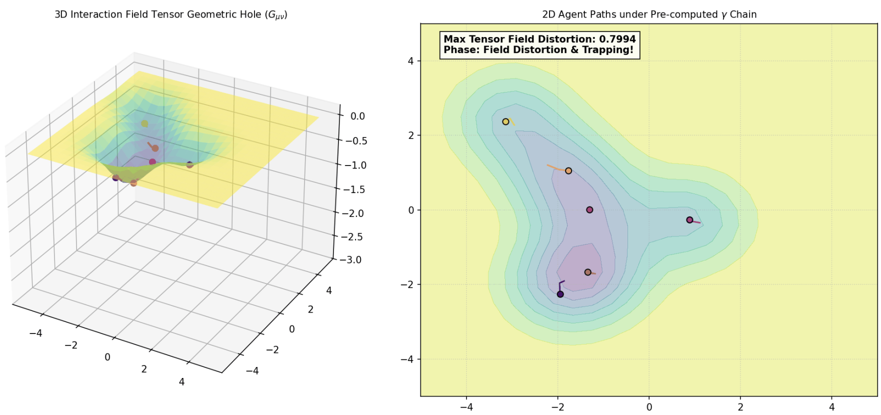


Figure 1: Static frame from the simulation.

The simulation produces two synchronised panels: a 3D surface plot of the interaction field landscape $Z(x, y, t)$ and a 2D top-view of agent trajectories under geodesic flow.

The key observable is the *maximum tensor field distortion* $\Delta_{\max} = |\min Z - \Lambda|$, which measures the depth of the curvature well. When $\Delta_{\max} > 0.3$, the system enters the "field distortion and trapping" phase: agent paths curve toward the curvature minimum, consistent with geodesic attraction on a negatively curved manifold.

This behaviour directly validates the interaction field equation (29): the aggregated source tensor $T_{\mu\nu}$ (34) deforms the interaction metric, generating curvature that in turn exerts geodesic forces on the agents— the geometric analogue of gravitational attraction in general relativity.

The noise term $\boldsymbol{\eta}_i$ represents behavioural stochasticity (the "emotional volatility" identified in the continuous layer), while the geodesic force (33) represents the deterministic structural coupling encoded in the Christoffel symbols (23).

# 10 Discussion

## 10.1 The Unified Structure

This paper has shown that four apparently distinct levels of description—discrete recurrence, continuous flow, emotional acceleration, and Riemannian field theory—are not separate theories but successive refinements of a single mathematical structure rooted in the Weber–Fechner psychophysical law.

Three conceptual implications stand out:

1. **Interaction is a dynamical process, not a static record.** The equation $\dot{I}_S = MI_S$ asserts that affect evolves continuously, like motion in physics, governed by intrinsic rates of change. This shifts the epistemology of emotional science from measurement to mechanism.

2. **Relationship behaviour is a spectral and geometric phenomenon.** Eigenvalues of $M$ determine convergence, escalation, or oscillation, while curvature of $\mathcal{M}$ encodes how the coupling $\gamma$ deforms interaction space. Cyclical emotional patterns arise when cross-agent influence exceeds stability thresholds—structurally, not circumstantially.

3. **Classical behavioural phenomena emerge as mathematical corollaries.** Thorndike's learning curve (9), Gottman's dyadic model, and the present geometric field theory are not competing descriptions but limiting cases of the same underlying dynamical principle.

## 10.2 The Role of the Field Equations

The interaction field equations (29) provide a geometric language for phenomena that have previously been described only qualitatively:

- $T_{\mu\nu} = 0$ (no reinforcement): flat interaction geometry, neutral dynamics.

- $T_{\mu\nu}$ positive definite (positive reinforcement): negative curvature, geodesic attraction, cohesive interaction.

- $T_{\mu\nu}$ indefinite (conflicted reinforcement): mixed curvature, non-trivial geodesic structure, potential for oscillatory or chaotic trajectories.

- $\Lambda > 0$: baseline repulsion at large interaction distances, analogous to cosmological expansion.

### 10.3 Clarifications and Limitations

Several important qualifications apply. First, the model does not claim to explain subjective emotional experience, meaning, or narrative cognition. Interaction variables are abstract mathematical quantities. Second, the equivalence with Gottman's model is structural, not epistemological; the present framework does not replace empirical relationship research. Third, the restriction to three-dimensional interaction space (one time, two agents) is a modelling choice; the framework extends to $n$ agents on $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^n_{I_S}$ straightforwardly.

# 11 Conclusion

We have developed, in full mathematical detail, a unified four-layer framework for psychophysical interaction dynamics:

1. **Discrete**: $I_S(t+1) = \gamma I_S(t)$ with Jury stability and bifurcation theory.
2. **Continuous**: $\dot{I}_S = M(t) I_S$ with generator equivalence $\gamma = e^{M\Delta t}$ and spectral stability.
3. **Emotional**: $\ddot{I}_S = E(t) I_S$, $E = \dot{M} + M^2$, with unique relaxation solution $M(t) = (t+\tau_0)^{-1}$ and healing time $T_{\text{heal}} \sim \mathcal{O}(\epsilon^{-1/2})$.
4. **Geometric field theory**: Riemannian manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ with metric $G = I + J^\top J$, Ricci curvature decomposition, three-layer geometric lifting, and Einstein-type field equation $R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$.

The numerical simulation validates the field theory: symbolic pre-computation of the full Riemannian structure (Christoffel symbols through Einstein tensor) followed by high-speed NumPy evaluation produces multi-agent geodesic dynamics consistent with the geometric prediction of curvature-driven attraction and trapping.

This framework establishes a mathematically rigorous and internally consistent foundation for the geometric study of interpersonal interaction, opening pathways toward curvature-based stability analysis, empirical parameter learning, stochastic field theory on interaction manifolds, and—ultimately—a fully nonlinear theory of emotional dynamics.

# Future Work

1. **Nonlinear emotional turbulence**: Riccati-type extensions of the emotion equation with saturation and thresholding.
2. **Stochastic interaction fields**: Itô diffusion on $\mathcal{M}$, modelling emotional volatility via $dM = f(M)\,dt + \sigma\, dW_t$.
3. **Higher-dimensional manifolds**: Extension to $n$-agent systems on $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^n_{I_S}$.
4. **Empirical parameter inference**: Learning $\gamma$ or $M$ from observational dyadic recordings; testing curvature-based stability predictions.

5. **Matrix Ricci flow**: Exploring connections between the evolution of $G(t, I_S)$ and Perelman–Hamilton Ricci flow on interaction manifolds.
6. **Explicit $T_{\mu\nu}$ models**: Specifying interaction source tensors for cognitive, social, and neurobiological systems.

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